

Future Improvements & Strategic Roadmap

This roadmap outlines high-impact architectural enhancements planned for subsequent versions of the AI Study Companion.

1. Advanced Retrieval & Search Enhancements

- **Hybrid Sparse-Dense Retrieval (BM25 + pgvector):** Combine traditional lexical search (BM25) with dense semantic embeddings via Reciprocal Rank Fusion (RRF) to excel at both conceptual queries and exact symbol lookups.
- **Cross-Encoder Re-ranking:** Add a second-stage re-ranking pass (e.g., Cohere Rerank or BGE-Reranker) to evaluate the top 20 candidate chunks before prompt assembly, boosting citation precision.
- **Hierarchical Document Trees:** Implement Parent-Document retrieval where small chunks are indexed for high similarity matching, but their larger parent sections are passed to the LLM for broader contextual grounding.

2. Spaced Repetition & Adaptive Scheduling

- **SuperMemo (SM-2) / FSRS Flashcards:** Automatically distill key concepts into digital flashcards with adaptive interval scheduling (review in 1 day, 3 days, 1 week, 1 month) based on recall difficulty.
- **Calendar & Learning Habit Integration:** Sync recommended review sessions with Google Calendar / Outlook to nudge learners when concepts show declining mastery trends.

3. Real-Time Collaboration & Multimodality

- **Shared Spaces & Peer Study:** Enable collaborative study spaces with multi-user permissions, shared document annotations, and group quiz leaderboards.
- **Voice-First AI Tutor:** Integrate real-time audio input/output (e.g., WebRTC + Whisper/ElevenLabs or Gemini Live) for conversational, hands-free study sessions while commuting or exercising.
- **Multimodal Visual Grounding:** Allow students to upload diagrams, architectural charts, and handwritten problem sets using vision models (`claude-3-5-sonnet` / `gemini-1.5-pro`) to

critique visual sketches.

4. Performance & Cost Optimizations

- **Anthropic Prompt Caching:** Cache unchanging document context blocks and system instructions across tutor messages to cut response latency by 50% and token costs by up to 90%.
- **Streaming WebSockets:** Transition from HTTP polling to WebSockets for instant document processing status, background recommendation notifications, and real-time collaborative activity.